Linear Programming based Financial Modelling approach for Optimal Net Interest Income

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ABSTRACT: Financial modelling is considered as one of the predominant skill having immense potential for growth optimization in financial, banking, industries and corporations. Here, linear programming approach is employed for financial modelling with the goal to optimize Net Interest Income (NII) that plays vital role in capital management and decision processes. Considering its significance towards deciding certain optimal profit on investments. We have attempted a Linear Programming to optimize NII with the constraints being a minimum cash balance for each month, the monthly lending being less than the monthly cash receipts & a non-negativity constraints for all lending, borrowings, purchase & collection fractions involved in the formulation. The decision variables being considered are monthly lending & borrowings as well as the 2-month collection & purchase fraction for these lending & borrowings. The financial inputs have been retrieved from the projected income statements, cash budget and balance sheet.

Keywords: *Financial Model, Linear programming, net interest income*

1. INTRODUCTION

In the present day scenario, the global economic crisis depreciates the business environment and makes it highly intricate for management. In such circumstances, the organization needs to learn and adapt the variations to ensure efficient and promising organizational growth. The administrators and managers must pay higher attention towards those paradigms, which would assist the company not only to endure but also to accomplish all expected objectives of future financial planning. On the other hand, the high pace change in business environment scenario has caused the emergence of the economy of higher dimensions. In such circumstances, the only thing that retains its stability is the optimal exploitation of its resources to continue business activities. In order to accomplish the optimistic forecast of the future decision process in financial activities, the paradigm of financial modelling plays the significant role. To incorporate the financial analysis for optimizing optimal growth, the mathematical and statistical approaches can be potentially implemented with automated Excel. Financial modelling can play a significant role in ensuring secure investments and optimal future planning, especially for financial analysis and decision process. Financial modeling states for the approach of creating financial representation of certain company so as to secure it with overall organizational aspects. In general it is accomplished by exhibiting optimistic evaluations and introducing optimal approach on the basis of information retrieved. Additionally, it should also facilitate specific events and recommendation for possible alternatives and profit oriented actions. Financial management models perform significantly for enabling business operations and risk estimations in optimal and cost-effective way against varied hypothetical assumptions, evaluation of the optimal solutions for calculating ensured financial returns with optimized business decision process. Especially in case of secure and optimal financial investment procedures, the estimation of net interest income (NII) can be well retrieved using financial modelling approach.

Financial modeling (Terry and Zeitlow, 2000) is a statistical paradigm representing the study on the inter-relationship among the variables in certain specific financial issues. The procedure of financial modeling

encompasses most of the inter-parametric analysis in financial terms so as to ensure precise outcome which guarantees the higher productivity and secure investment. Incorporating this paradigm, varying the input as well as output financial parameters the variation in the dependent variables could be visualized which assists for future financial planning and execution. In spite of this, it is expected that some of the inter-relationships are flexible to be configured where few are evaluated as the approximate while others even remain uncertain. Those could be retrieved by employing well rooted financial theory, statistical analysis approaches and resulting data analysis. Evaluation of these data becomes mammoth and intricate issue in financial modelling process because of the fact that the more inter-relations are known in modelling, better prediction could be achieved.

To assure optimal benefits and secure investment, there are certain parameters which are required to be optimized and Net Interest Income (NII) is the key factor amongst them. NII represents the difference between the income generated by certain interest bearing goods and the cost of servicing liabilities.

In general, the financial models are generated with an objective to analyze the approach in which the parametric values do change as per variation in the single or multiple independent variables. In this study NII is the dependent variables which have been projected and determined using developed financial modelling. Additionally, a number of auxiliary dependent variables have been obtained that could be employed for further analysis. In fact the process of financial modeling represents a systematic paradigm for financial study that comprises the perspective planning and varied other phases. The first phase of financial modelling is the structuring the financial issues where the problems and its probable projected approach of analysis are discussed. It is then followed by defining the input and output financial entities along with the incorporation of analytical algorithms so as to get more precise and accurate outputs. To formulate and design certain financial model a well skilled effort is needed which is enriched with optimistic vision about the expected or targeted outputs using statistical and mathematical modelling. In fact the functional approaches in the financial modelling depend on the characteristics of the specific financial issues where it would be advisable to model the fundamental model approach and adjust the variables as per the current situations. The functional as well as procedural flexibilities of the Excel make it the best choice to conduct financial modelling.

This manuscript discusses the financial modelling using linear programming (LP) approach with defined objective functions of Net Interest Income (NII). Here the study has been emphasized over optimization of organization decision process and capital management on the basis of NII. The presented model discusses optimization issues NII where the data analysis has been done with the input taken from projected income statement, projected cash budget and its projected balance sheet. In this study collection fraction, purchase fraction, minimum cash balance and ending inventory have been considered as policy variables. Two variables, borrowing and lending have been considered as decision variables. In order to analyze the impact of variation in objective functions, the policy variables can be varied. The financial data are from Prowess datasets and Excel SOLVER is employed for solving the data structure and optimization.

2. RELATED WORKS

Hillier and Lieberman (2001) stated that though resource allocation for varied business activities is generic application; the employment of LP could be playing much significant role for financial and resource planning optimization. Backing up this paradigm Dwivedi (2008) advocated that LP could be a significant approach for making optimal decision process in organizations and it could be helpful for evaluating various economic relations. It also strengthens the employability of LP for NII estimation using financial modelling. Other authors (Gupta et al, 2009) suggested that LP in general deals with linear optimization of certain objective function with linear equations or inequalities as constraints.

The LP scheme emphasizes over the objective functions, which are needed to be optimized, decision variables are expected to be estimated and the constraints are supposed to be satisfied. In some other researches, (Wagner, 2007, Lucey, 2002) LP has been suggested as an operational research scheme and LP was recommended as the most efficient scheme for decision process optimization. The author also presented that the LP model can be potentially adopted for financial modelling and management.

Susilawati et al (2002) employed LP for premise resource optimization using SOLVER. The effectiveness of LP for manpower planning was done by Kareem et al (2008) in a cocoa processing industry in Akure, Ondo State of Nigeria. In another work Murugan et al, (2009) employed the LP scheme with a software application called linear interactive optimizer (LINDO) to maximize the organizational profits.

Jiang et al. (2011) advocated the cash flow at the project level comprising a comprehensive history of all cash expenditure, cash shortage, loans, cost of money, and all incomes received as a result of project execution. El-Khol, (2014) developed a multi-objective fuzzy linear programming model for cash flow management. The author considered the optimization issue with three predominant conflicting objectives, cost of money, final cash balance, and initial cash balance. A LP based portfolio optimization scheme was proposed by Mansini et al, (2014) where the authors employed mixed integer linear programming (MILP) models. Taking into consideration of the above mentioned reviews it can be concluded that LP schemes can play significant role in asserting efficient optimization for financial modelling resulting into precise and more productive decision process. Thus, getting motivation from these factors in this paper, LP scheme has been employed for NII optimization for financial modelling.

3. LP ENRICHED FINANCIAL MODELLING

In the presented study, the linear programming scheme has been employed for Net Interest Income (NII) optimization in financial modelling with a goal to ensure higher profits on investment. Here, the Linear Programming paradigm has been employed to evaluate the optimal objective function (NII) and the inter-dependent influences on decision process. To perform the analysis, the financial data has been retrieved from Prowess datasets. The LP enriched optimization has been employed for NII individually with certain defined constraints. Other dominant comprising elements under consideration are the LP approach of generic form with linear objective function, a set of linear inequality constraints and non-negative constraints for financial analysis. In this study, the projected income statement encompasses key financial parameters such as, sales revenue cost of goods, gross profit on sold goods, general and administrative depreciation, operating profit, interest income, and profit before tax, taxes and ultimate net income. The policy variables are collection fraction, purchase fraction, minimum cash balance and ending inventory. The decision variables are borrowing and lending. The policy variables are varied for various fractions to identify the impact on the objective functions. Taking into account of this parameter (NII) the LP scheme has been employed to achieve optimal values to ensure better profits and secure investment. In the presented study, the optimization scheme LP has been employed for objective functions NII to evaluate the effects of dependent variables in financial structures. A generic approach of Linear Programming implementation for NII optimization has been illustrated as follows:

3.1 Net Interest Income as Objective Function

In the first phase of LP implementation for financial modelling, NII has been considered as the objective function. Consider X_i be the Objective functions of the values $A_1, A_2, \ldots, A_m$ where $X_i \in N$ and $i \in 1, 2, \ldots m$, whereas the individual entity X_i is comprised of numerous financial entities and its combination to achieve financial models S_r, where $r \in 1, 2, \ldots, n$. Now, for financial modelling it is needed to define the objective functions and varied conditions representing the constraints because of confined availability of every entities or substances. Here, NII has been considered as objective function to optimize interest income over investment by selling the services or goods. To illustrate the optimization process with maximization goal the following situation could be taken into consideration. The overall linear optimization could be illustrated mathematically as mentioned below.

$$MAX\ Z = [a_1 X_1 + a_2 X_2 + a_3 X_3 + a_4 X_4 + a_5 X_5 + a_6 X_6 + a_7 X_7 + a_8 X_8 + a_9 X_9 + a_{10} X_{10}] - [a_{11} X_{11} + a_{12} X_{12} + a_{13} X_{13} + a_{14} X_{14} + a_{15} X_{15} + a_{16} X_{16} + a_{17} X_{17} + a_{18} X_{18} + a_{19} X_{19} + a_{20} X_{20}]$$

Where the decision variables X_1 to X_{20} determine the volume of lending and borrowing for a given rate of interest

X_1= 30 day (lending) Cash Disbursement(CD) at 8% interest rate for first month
X_2= 30 day (lending) Cash Disbursement (CD) at 9% interest rate for second month
X_3= 30 day (lending) Cash Disbursement(CD) at 10% interest rate for third month
X_4= 30 day (lending) Cash Disbursement(CD) at 9.5% interest rate for fourth month
X_5= 30 day (lending) Cash Disbursement (CD) at 9% interest rate for fifth month
X_6= 30 day (lending) Cash Disbursement (CD) at 9% interest rate for sixth month
X_7= 90 day (lending) Cash Disbursement (CD) at 9.5% interest rate for first month
X_8= 90 day (lending) Cash Disbursement (CD) at 9% interest rate for second month
X_9= 90 day (lending) Cash Disbursement (CD) at 9.5% interest rate for third month
X_{10}= 90 day (lending) Cash Disbursement (CD) at 9% interest rate for fourth month
X_{11}= 30 day (borrowing) Notes Payable (NP) at 9% interest rate for first month
X_{12}= 30 day (borrowing) Notes Payable (NP) at 10% interest rate for second month
X_{13}= 30 day (borrowing) Notes Payable (NP) at 11.5% interest rate for third month
X_{14}= 30 day (borrowing) Notes Payable (NP) at 11% interest rate for fourth month
X_{15}= 30 day (borrowing) Notes Payable (NP) at 9.5% interest rate for fifth month
X_{16}= 90 day (borrowing) Notes Payable (NP) at 9.5% interest rate for first month
X_{17}= 90 day (borrowing) Notes Payable (NP) at 10% interest rate for second month
X_{18}= 90 day (borrowing) Notes Payable (NP) at 11% interest rate for third month
X_{20}= 180 day (borrowing) Notes Payable (NP) at 10% interest rate for first month

The Constraints used for the LPP formulation were as follows:

a) $0 \leq X_1 + X_7 \leq$ Total Cash Receipts for first month

b) $0 \leq X_2 + X_8 \leq$ Total Cash Receipts for second month

c) $0 \leq X_3 + X_9 \leq$ Total Cash Receipts for third month

d) $0 \leq X_4 + X_{10} \leq$ Total Cash Receipts for fourth month

e) $0 \leq X_5 \leq$ Total Cash Receipts for fifth month

f) $0 \leq X_6 \leq$ Total Cash Receipts for sixth month

Where Total Cash receipts for any month month= (Collection fraction for that month* Sales of that month + Collection fraction for 1 month after sale * Sales of previous month + Collection fraction for 2 months after sale * Sales of 2 months before that month) & so on.

g) Minimum Cash Balance for first month= Initial cash balance for first month + Total cash receipts for first month - Total cash disbursements for first month – $X_1 - X_7 + X_{11} + X_{15} + X_{19}$- Long Term Debt

h) Minimum Cash Balance for second month = Initial cash balance for second month + Total cash receipts for second month - Total cash disbursements for second month + $X_1 – X_2 – X_8 + a_1X_1 + a_7X_7 - a_{11}X_{11} – a_{16}X_{16} - a_{20}X_{20}$ - Long Term Debt

i) Minimum Cash Balance for third month = Initial cash balance for third month + Total cash receipts for third month - Total cash disbursements for third month + $X_2 – X_3 – X_9 + a_2X_2 + a_7X_7 + a_8X_8 - a_{12}X_{12} - a_{16}X_{16} - a_{17}X_{17} - a_{18}X_{18} - a_{20}X_{20} - X_{12} + X_{13} + X_{17}$ - Long Term Debt

j) Minimum Cash Balance for fourth month = Initial cash balance for fourth month + Total cash receipts for fourth month - Total cash disbursements for fourth month + $X_3 – X_4 + X_7 – X_{10} + a_3X_3 + a_7X_7 + a_8X_8 + a_9X_9 - a_{13}X_{13} - a_{17}X_{17} - a_{18}X_{18} - a_{20}X_{20} - X_{13} + X_{14} - X_{16} + X_{19}$ - Long Term Debt

k) Minimum Cash Balance for fifth month = Initial cash balance for fifth month + Total cash receipts for fifth month - Total cash disbursements for fifth month + $X_4 - X_5 + X_8 + a_4X_4 + a_8X_8 + a_9X_9 + a_{10}X_{10} - a_{14}X_{14} - a_{17}X_{17} - a_{18}X_{18} - a_{19}X_{19} - a_{20}X_{20} - X_{14} + X_{15} - X_{17}$ - Long Term Debt

l) Minimum Cash Balance for sixth month = Initial cash balance for sixth month + Total cash receipts for sixth month - Total cash disbursements for sixth month + $X_5 - X_6 + X_9 + a_5X_5 + a_9X_9 + a_{10}X_{10} - a_{15}X_{15} - a_{18}X_{18} - a_{19}X_{19} - a_{20}X_{20} - X_{15} - X_{18}$ - Long Term Debt

m) Minimum Cash Balance ≥ 0

n) 0≤collection fraction % for month of sale <=100

o) 0≤ collection fraction % for 1 month after sale≤100

p) 0≤ collection fraction % for 2 months after sale≤100

q) 0≤ purchase fraction % for month of sale≤100

r) 0≤ purchase fraction % for 1 month after sale≤100

s) $X_i \geq 0$

The different kinds of variables that has been employed in the LPP formulation as follows:

3.1 Policy Variables

a) Collection fraction for current month of sale

b) Collection fraction for 1 month after sale

c) Collection fraction for 2 months after sale

d) Purchase fraction for current month of sale

e) Purchase fraction for 1 month after sale

f) Minimum Cash Balance

ILLUSTRATION

Now that we have assembled both the development as well as the optimization part of our financial planning model we are ready to test it with a known sample problem to ensure its credibility.

The sample problem is as follows,

Program a financial planning model using the sales data & the tabular information provided in tables 1 through 3. Put zero for borrowings/debt and lending/loan variables. Assume that projected sale under the term are the following:

January	*Rs 1700*	*April*	*Rs 2600*
February	*Rs 1800*	*May*	*Rs 2400*
March	*Rs 2000*	*June*	*Rs 1900*

The company has specified a plan that results in a net interest of Rs. -99.83 earned for the six month planning period and requires the use of 180 days notes payable financing commencing in January at an interest rate of 10%, 90 days notes payable financing commencing in February at an interest rate of 10 % and 30 days notes in the month of May at an interest rate of 9.5%. Further the company has decided to use a collection fraction such that 10% of the sales is taken from the month in which sales take place, 10% in the next month and the remaining 80% of the sales is taken during the 2nd month after the sales has occurred. How much does the proposed credit policy differ from an optimal one? Would a change in credit policy be advantageous to the firm? (Terry and Zeitlow, 2000 Edn)

Also the tax payment months are on March 3rd, June 6th & December 12th.The tax rate is 4% and the rate on existing Long term debt is 10.5%.

Refer the table 1 for base period and sales forecast information. The table provides the input for the problem.

Table 1: Base period sales forecast information

	Base Period Information and Sales Forecast								
	Novembe	December	January	February	March	April	May	June	July
Sales	1400	1500							
Operating Expenses			250	250	250	250	250	250	
Depreciation								100	
Cash		250							
ST Investments		0							
Accounts Recievable		1770							
Inventory		1275							
Gross fixed Assets		4000							
Accum. Depreciation		1200							
Accounts Payable		1275							
Notes Payable		0							
Taxes Payable		200							
Long erm Debt		1000							
Common Stock		1000							
Retained Earnings		2620							

Refer the table 2 below for the types of debt and loan instruments as well as their expected term structure.

Table 2 : Types of debt and load instruments

	Expected Term Structure of Interest					
	Jan	Feb	Mar	Apr	May	Jun
Investment Instruments						
30 day Investments	8	9	10	9.5	9	9
90 day Investments	9.5	9	9.5	9	9.5	9.5
Financing Instruments						
30 day Notes Payable	9	10	11.5	11	9.5	9.5
90 day Notes Payable	9.5	10	11	11	10	10
180 day Notes Payable	10	10	10	10	10.5	10.5
Long Term Debt	10.5	10	10	10	11	11

Refer the table 3 for information regarding the policy variables that are to be used for the initial part of the problem.

Table 3 : Policy Variables

	Policy Variables		
	Month of	1 Month after	2 Months after
Collection fraction	10%	10%	80.0%
	Month of	1 Month after	
Payment Fraction	0%	100%	
Inventory Policy:	Ending Inventory =Next Months COGS		
Gross Profit Margin	25%		
Minimum Cash Balance	250		
General	250		
and Administrative Depreciation	0		

4.1 Solution of the Problem

Firstly we validate the model that we are preparing with respect to the company specified loan and debt instruments as well as the company specified policy variables i.e. collection fraction. On running the solver we obtain an NII of -98.0 (Figure 3) which closely equates to the company's NII of 99.83, thus validating the model we have prepared.

Maximize	
Z	-98.01855705

Figure 3 The Objective function for the sample problem after running the Solver

DECISION VARIABLES		
X1	30 day @0.08 (CD) Jan	0
X2	30 day @0.09 (CD) Feb	462.4997773
X3	30 day @0.1 (CD) Mar	0
X4	30 day @0.095 (CD) Apr	1900
X5	30 day @0.09 (CD) May	0
X6	30 day @0.09 (CD) Jun	0
X7	90 day @0.095 (CD) Jan	1490
X8	90 day @0.09 (CD) Feb	0
X9	90 day @0.095 (CD) Mar	0
X10	90 day @0.09(CD) Apr	0
X11	30 day @0.09 (N/P) Jan	1144.26182
X12	30 day @0.1 (N/P) Feb	0
X13	30 day @0.115 (N/P) Mar	-5.68434E-14
X14	30 day @0.11 (N/P) Apr	0
X15	30 day @0.095 (N/P) May	488.3060431
X16	90 day @0.095 (N/P) Jan	0
X17	90 day @0.1 (N/P) Feb	2415.543462
X18	90 day @0.11 (N/P) Mar	-5.68434E-14
X19	90 day @.11(N/P) Apr	729.3229539
X20	180 day @.1(N/P) Jan	389.4881803

Figure 4 The decision variables for the sample problem after running the Solver

The various decision variables obtained after running solver are shown in figure 4. Here the table is split into 3 columns; the first column shows the symbols used to depict the decision variables; the second column shows what the decision variable represents and gives us information as to how long and at what rate the amount was borrowed (Notes Payable or N/P) or lend (Cash Disbursement or CD), also when it was borrowed or lend and the final column shows the value it takes. The last column shows us the value of the decision variable.

Further on inspection we see that our decision variables (borrowings/lending) are different from the company's proposed variables. This clearly indicates that there exist multiple solutions for the same NII.

GRAPHICAL INFERENCE

The model validation helps us understand the fact that the decision variables used by the company were indeed optimal. However we are yet to see whether the companies proposed policy variables are truly optimal or not. To evaluate this we make use of graphical patterns to reveal to us how the NII behaves with respect to different policy variables. The inferences for the graphs are made on account of the knowledge of binding constraints.

All the three graphs as seen below show variation of NII with respect to a change in collection fraction for the three periods i.e. for the current month of sale, for 1 month after sale & for 2 months after sale. The collection fraction started from 10% during the first month, 60% during the second month and 30% during the third month). We vary two of these collection fractions keeping the other one constant. Also we make it a point to increment the collection fractions by margins of 5%.

The figure 5 shows how the NII varies when the collection fraction for the current month of sale is kept constant at 0% and the other collection fractions i.e. collection fractions for 1 month after sale & 2 months after sale is varied by a margin of 5%.

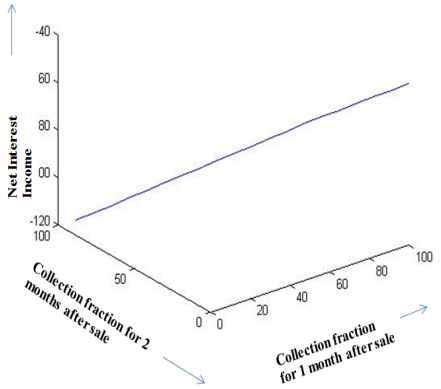


Fig 5 NII Vs CF for 2 months after sale Vs CF for 1 month after sale

The figure 6 shows how the NII varies when the collection fraction for 1month after sale is kept constant at 0% and the other collection fractions i.e. collection fractions for the current month of sale & 2 months after sale is varied by a margin of 5%.

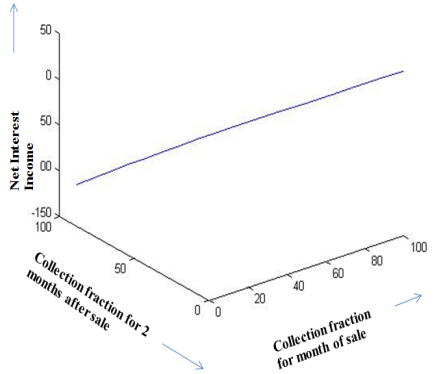


Fig 6 NII Vs CF for 2 months after sale Vs CF for month of sale

The figure 7 shows how the NII how the NII varies when the collection fraction for 2 months after sale is kept constant at 0% and the other collection fractions i.e. collection fractions for the current month of sale & 1 month after sale is varied by a margin of 5%.

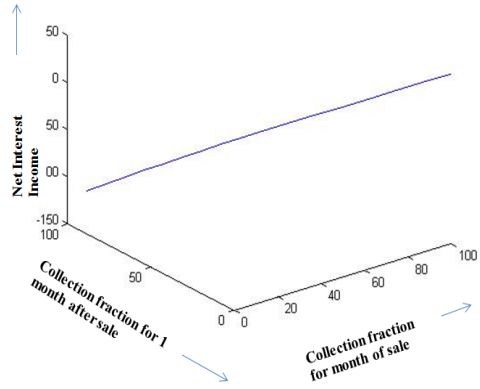


Fig 7 NII Vs CF for 1 month after sale Vs CF for month of sale

We can infer from all three graphs that the higher the percentage of collection from the present term/period or month of consideration (i.e. the faster we collect the cash) the more will be the net interest income. So from the graph we conclude that the highest net interest that can be collected for our presented problem is Rs.30.615 which is obtained when 100% collection fraction is employed during the month of sale.

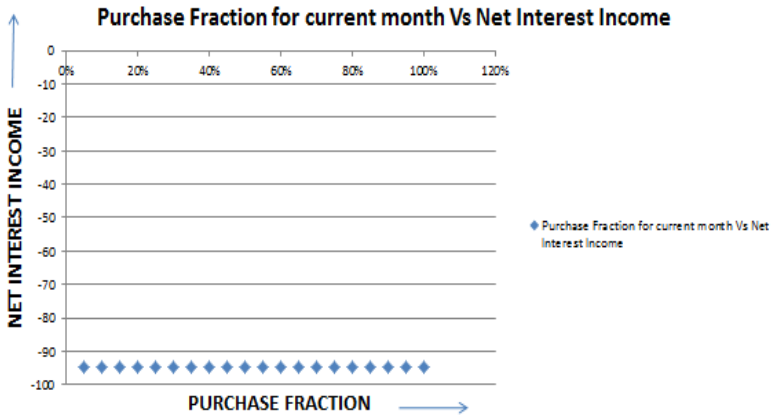


Fig 8 NII Vs PF

The figure 8 shows the variation of NII with respect to purchase fraction. As can be seen there is no variation in NII with changes in purchase fraction for current and next month, which reveals the fact that purchase is a non-binding constraint

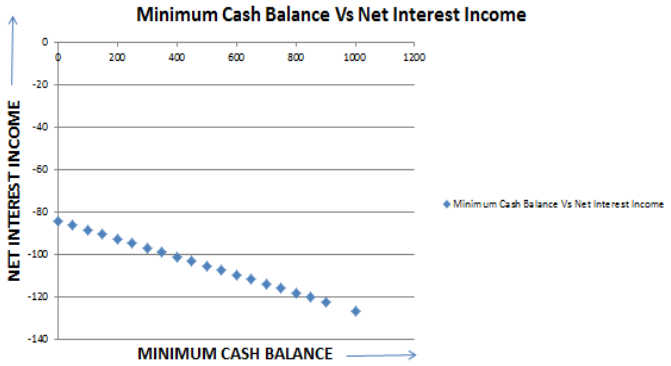


Fig 9 NII Vs Minimum Cash Balance

The figure 9 shows that purchase fraction is a binding constraint. On varying the minimum cash balance it can be seen that the graph has a downward slope. So the maximum value of the NII (negative as seen in the graph) is when zero minimum cash balance is maintained.

Hence from all the graphs that we have plotted our primary concern would be those variables which are the actual binding constraints. So it is to be understood that the collection fraction and minimum cash balance does impact the NII and hence should be kept at appropriate values that maximizes the NII.

1. CONCLUSION

Considering the requirement of an efficient and robust financial modelling scheme for organizational decision process, here in this paper a novel scheme has been proposed with enhanced LP based optimization capability. In this study a key financial factor, Net Interest Income has been optimized using LP optimization paradigm so as to

maintain profits as well as security on investments. The financial database, Prowess has been used to obtain the data and Linear Programming is used for optimizing NII, to solve the LPP model.

The graphical results obtained from the LPP formulation clearly contradicts the results obtained using the problem specified collection fraction percentages and the minimum cash balance.

The LPP formulation reveals to us that in order to obtain the highest Net Interest Income one should effectively maintain a minimum cash balance of Rs. 0 while holding a collection fraction of 100% during the month of sale, whereas the problem had recommended to use a minimum cash balance of Rs 250 and a collection fraction of 80% for month of sale, 10% for 1 month after sale, 10% for 2 months after sale.

Ultimately, the study concludes that the developed financial modelling scheme would help a lot in understanding the characteristics that how every financial decisions affects other inter-dependent parameters in financial and hence organization decision process. The optimization of NII with linear programming approach could be significant to ensure highly precise and accurate decision making process in financial institutions and thus the investment could be ensured for its security and benefits. Analyzing NII responses the financial policies could be rectified to get higher benefits on investments and optimal working environment.

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